# AMRANE ANES


AI Engineer | Machine Learning | Full-Stack Developer

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## SUMMARY


AI Engineering graduate specialising in machine learning, computer vision, nlp and graph neural networks. Built and deployed an end-to-end multimodal AI pipeline for solar panel defect detection and maintenance recommendation as a graduation project, covering deep learning, model fine-tuning, knowledge graph construction, and RGAT-based classification. Complementary full-stack development background with production web and NLP projects. Seeking an AI/ML or software engineering role with an international or remote-friendly team.


## EDUCATION

**Master's in Intelligent Systems Engineering (AI)** | Université Saad Dahleb Blida

*September 2024 – June 2026 | Algeria*

- Graduation project: AI-Based Predictive Maintenance for Solar Panels Using Multimodal Knowledge Graph.
- Implemented EfficientNet-B0 fine-tuning, preprocessing (CLAHE), MMKG construction linking visual features and structured metadata, and Relational Graph Attention Network (RGAT) training for fault classification.
- Core coursework: Deep Learning, Graph Neural Networks, Computer Vision, Intelligent Agents & Multi-Agent Systems.

**Bachelor's in Computer Science** | Université Saad Dahleb Blida

*Graduated 2024 | Algeria*

- Final-year project: Tabibi — full-stack web platform for online medical appointment booking.

## TECHNICAL SKILLS

**AI / ML:** PyTorch, PyTorch Geometric, EfficientNet, RGAT, Scikit-learn, spaCy, OpenCV, NumPy, Pandas

**Knowledge Graphs & GNNs:** MMKG construction, graph attention networks (GAT / RGAT), property graphs

**NLP:** Text classification, sentiment analysis, abusive speech detection & rewriting, Arabic NLP

**Programming Languages:** Python, JavaScript, TypeScript

**Web & Backend:** Next.js, React, Node.js, Express.js, MySQL, Tailwind CSS, REST APIs

**Tools & DevOps:** Git, GitHub, Docker (basics), Vercel, n8n (workflow automation), VS Code

## PROJECTS

**AI-Based Predictive Maintenance for Solar Panels Using Multimodal Knowledge Graph** | Master's Graduation Project defended with Excellence

*Juin 2026*

- Built an end-to-end multimodal pipeline (EL image preprocessing, EfficientNet-B0 fine-tuning, N-MMKG construction in PyTorch Geometric, RGAT training) for 5-class solar cell defect classification across 15,900 cells / 265 panels.
- Achieved F1-Macro of 0.8092, outperforming RGCN and RGAT+MLP baselines; ablation study showed visual (EL image) modality is the primary discriminative signal (F1 drops to 0.21 without it).
- Deployed as an interactive Streamlit inference dashboard; incubated as a startup (SolarGuard) at CDE Blida 1 under Ministerial Decree 1275.

**AI News Automation Agent** | Personal Project

*2026*

- Built a reactive agent pipeline in n8n: scrapes Hacker News AI threads, processes content via Groq LLM, and publishes French-language summaries to Discord — fully automated.
- Framed and documented the system using the Wooldridge & Jennings agent-property framework (reactivity, proactivity, social ability).

**Real Time Fire And Smoke Detection** | University Project

*2026*

- Built a YOLOv8-based real-time fire and smoke detection system.
- Achieved mAP@0.5 of 0.803 on the validation dataset.
- Applied data augmentation, dataset cleaning, explainable AI techniques, and model comparison experiments.

**Abusive-to-Friendly Speech Transformer** | NLP Project
*2025*

- Developed an NLP pipeline to detect hostile or abusive text and rewrite it in a neutral tone; supports Arabic and English.
- Includes a CLI sentiment analysis tool for both languages.

**PDF-Based AI Chatbot (RAG)** | Personal Project
*2025*

- Built a retrieval-augmented generation chatbot that ingests PDF documents and answers natural-language questions via an NLP pipeline (chunking, embedding, retrieval).

**Tabibi — Medical Appointment Platform** | Bachelor's Final-Year Project
*Juin 2024*

- Full-stack web application: Node.js / Express REST API, MySQL database, React / Tailwind CSS frontend.
- Features: doctor scheduling, patient records, appointment booking and management.

**Personal Portfolio** | anes-amrane.vercel.app
*2024*

- Next.js + TypeScript + Tailwind CSS, dark mode, GitHub API integration, blog, and downloadable resume.

## EXPERIENCE

Developed multiple end-to-end AI and software engineering projects, including computer vision, NLP, knowledge graphs, and full-stack applications.

## LANGUAGES

**Arabic:** Native
**English:** B2 — Professional working proficiency
**French:** B1 — Intermediate

## ADDITIONAL

- Open to relocation and fully remote positions; actively exploring international AI engineering opportunities.
- Consistent self-learner: built multiple production-quality projects independently alongside full-time academic work.
- Interests: generative AI, agent-based systems, graph neural networks, startup culture, machine learning and computer vision and nlp.